

SIMATS ENGINEERING ASSESSMENT TOOL 2

**COURSE NAME: CSA07 COURSE CODE: Computer
Networks**

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A CONCEPT MAPPING

Dr.V.PARTHIPAN

Code of Conduct:

I T.GANESH(192425246) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

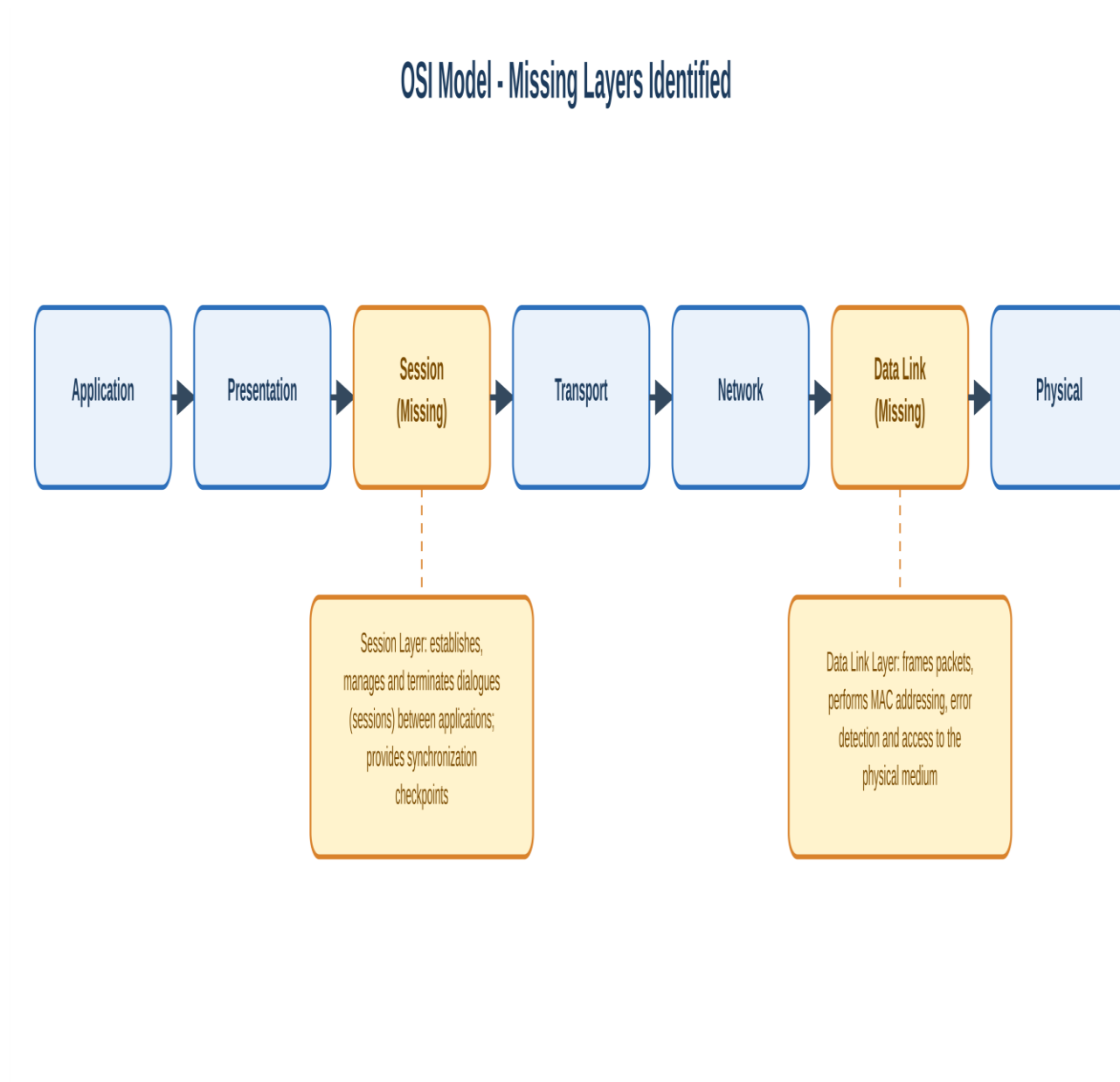
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

OSI Model — Concept Maps

Question 1

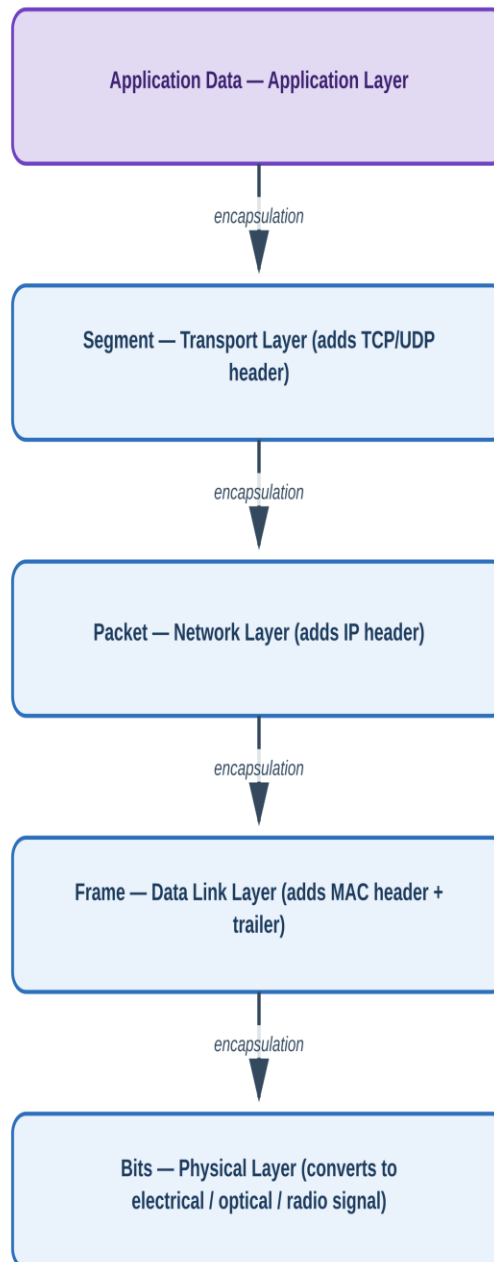
A partially completed concept map is given below: Application → Presentation → _____ → Transport → Network → _____ → Physical. Analyze the missing layers and explain how their functions are essential for reliable communication.



Question 2

Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

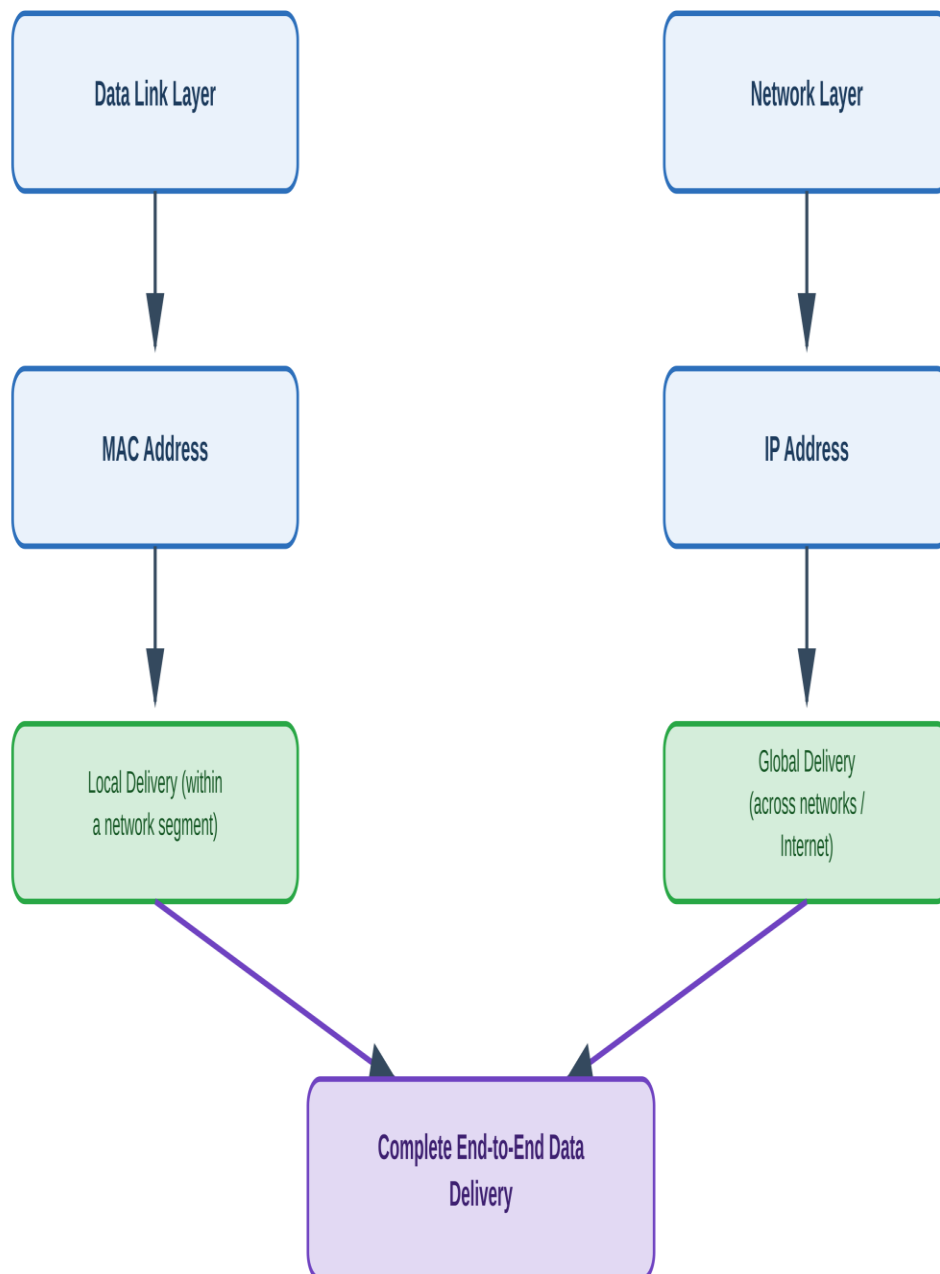
Data Units vs OSI Layers (Encapsulation)



Question 3

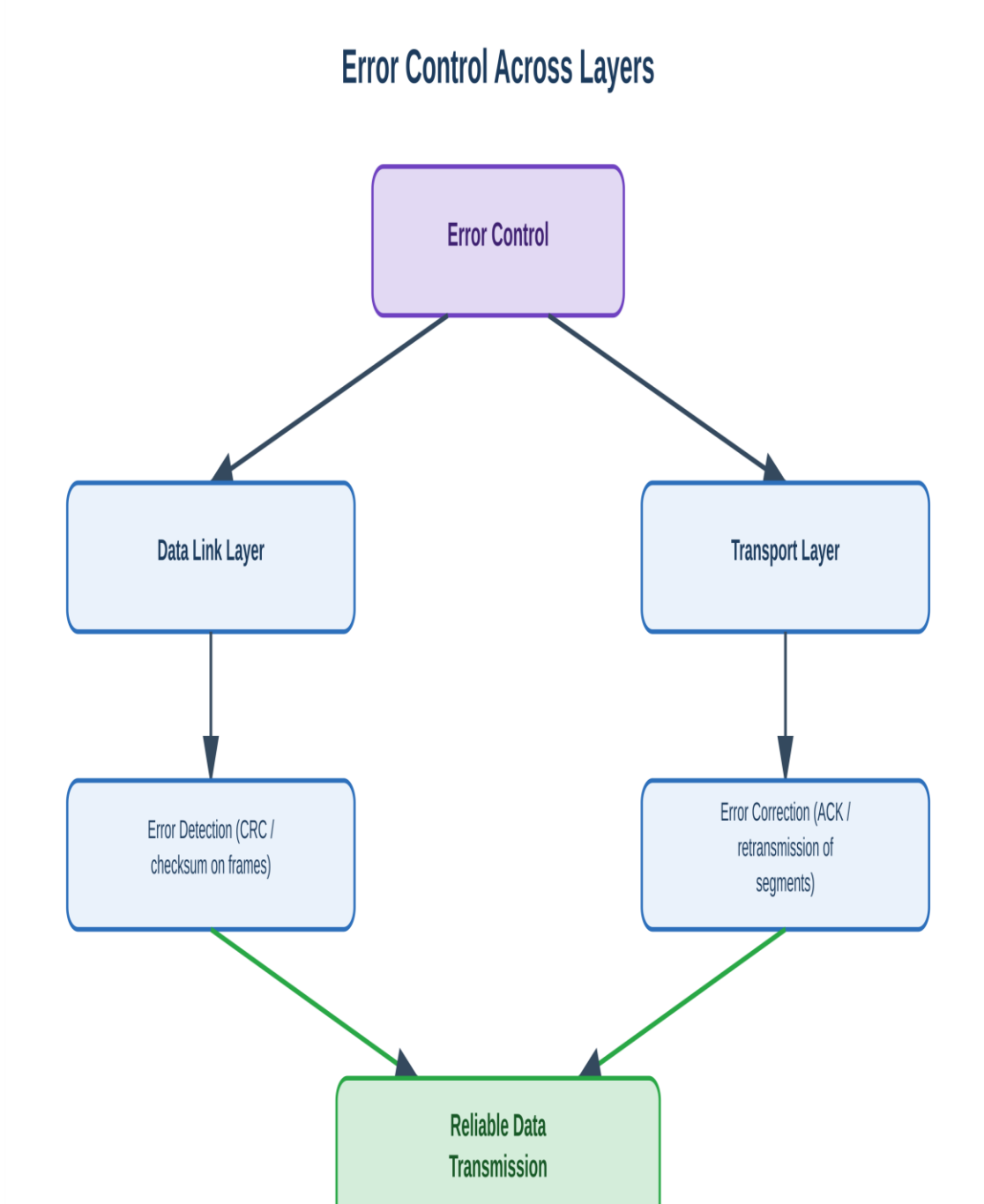
Given the concept map: DataLink → MAC Address → Local Delivery, Network → IP Address → Global Delivery. Explain how these relationships ensure complete data delivery across networks.

Local vs Global Delivery



Question 4

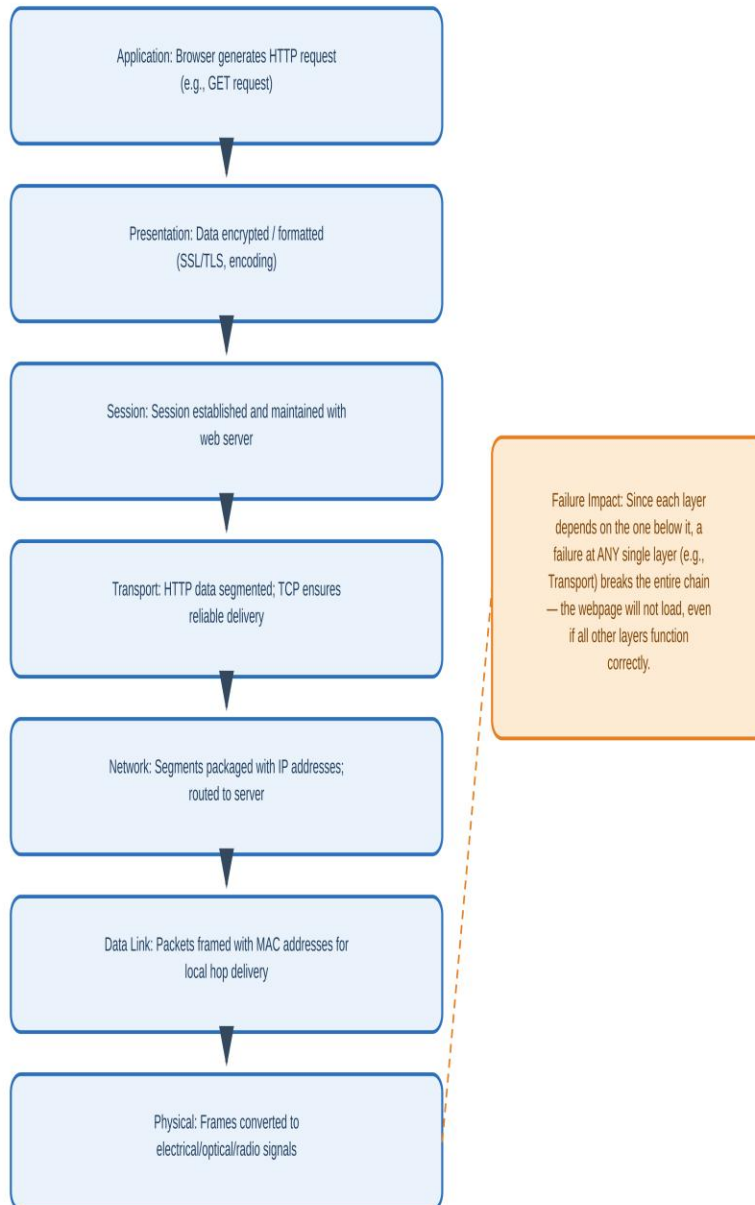
A concept map shows: Error Control → Data Link Layer → Error Detection, → Transport Layer → Error Correction. Analyze how the interaction between these layers improves reliability in data transmission.



Question 5

Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

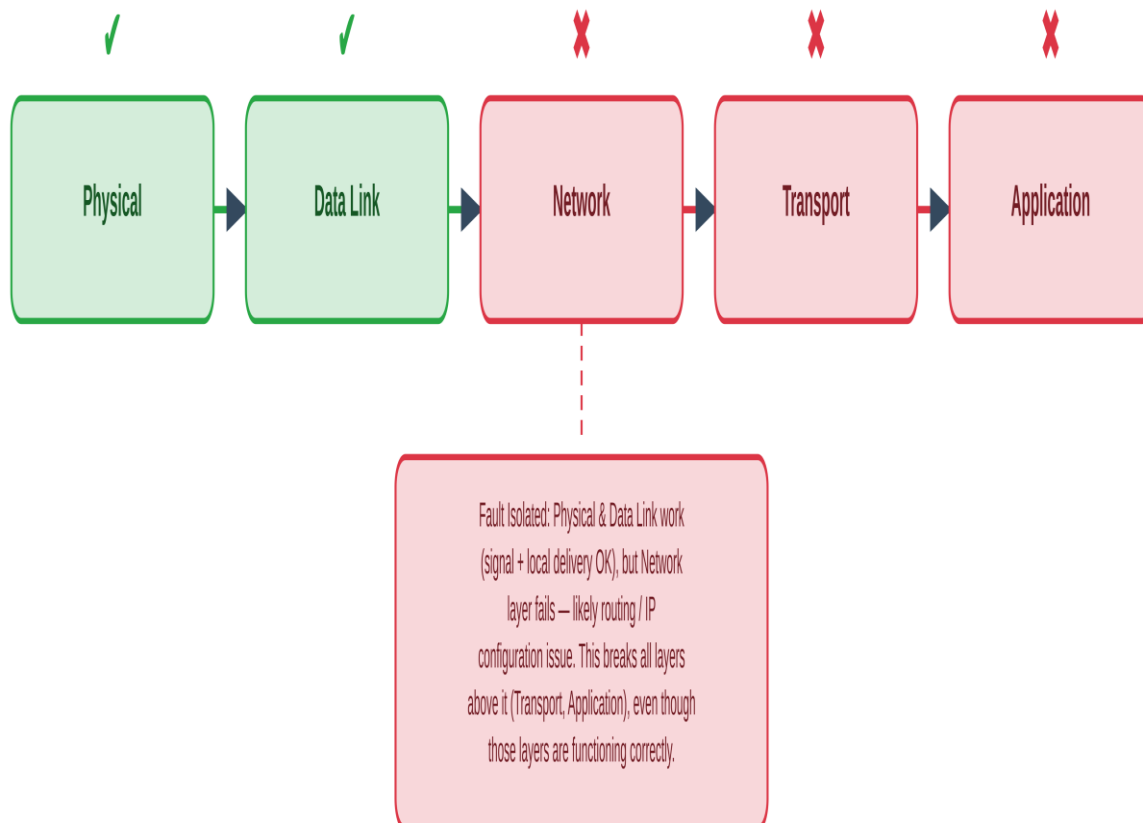
Web Browsing Through the OSI Layers



Question 6

A network issue is represented in the concept map: Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗. Analyze the problem and explain how the concept map helps in troubleshooting.

Network Fault Isolation Using a Concept Map



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers